

PRINCE SINGH

RAG PORTFOLIO KNOWLEDGE BASE — v2

Expanded from historical resumes + current public profile evidence + verified public repository information

Name	Prince Singh
Degree	B.Tech — Computer Science & Engineering (Data Science)
College	G.L. Bajaj Institute of Technology & Management, Greater Noida
Batch	2027
Current resume CGPA	8.0/10
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Phone	+91 9560730326
Portfolio	https://vibieprince.vercel.app
GitHub	https://github.com/vibieprince
LinkedIn	https://www.linkedin.com/in/vibieprince
LeetCode	https://leetcode.com/u/vibieprince/
Instagram	https://www.instagram.com/vibieprince

CONTENTS

- 01. 01 Identity & Professional Summary
- 02. 02 Education
- 03. 03 Master Skills Inventory
- 04. 04 Programming & Core CS
- 05. 05 Data Science & Analytics
- 06. 06 Web & Backend
- 07. 07 Databases
- 08. 08 Cloud / DevOps / Data Engineering
- 09. 09 AI / ML / GenAI
- 10. 10 Project Registry — with Deployment Links
- 11. 11 Hackathons & Competitive Builds
- 12. 12 Achievements
- 13. 13 Certifications — consolidated across resumes + public profile
- 14. 14 Experience / Practical Engineering
- 15. 15 Leadership / Collaboration
- 16. 16 GitHub / Open Source
- 17. 17 LeetCode / CodeChef / HackerRank
- 18. 18 Career Interests
- 19. 19 Social / Contact / Professional Links
- 20. 20 FAQ
- 21. 21 RAG Grounding / Source Rules
- 22. 22 Verification & Missing-Data Register

01. IDENTITY & PROFESSIONAL SUMMARY

Prince Singh is a B.Tech Computer Science & Engineering (Data Science) student at G.L. Bajaj Institute of Technology & Management, Greater Noida, in the 2023–2027 batch.

His profile combines software/backend engineering, data science and analytics, cloud technologies, AI/ML and generative-AI application development. Recurring technologies include C++, C, Python, JavaScript, SQL, FastAPI, REST APIs, Firebase, PostgreSQL, MySQL, MongoDB, Redis, Docker, Linux, AWS, GCP, Power BI, Snowflake, dbt, LangChain, ChromaDB, Gemini and Mistral.

He has explicitly confirmed that he has learned Linux commands and has worked with AWS and Google Cloud. These are therefore treated as confirmed profile facts in this v2.

Career interests: software engineering/backend, cloud/DevOps, data science/data engineering and AI/GenAI engineering. He prefers practical projects that move from problem definition to implementation, debugging and deployment.

02. EDUCATION

G.L. Bajaj Institute of Technology & Management, Greater Noida — B.Tech CSE (Data Science), 2023–2027. Current resume snapshot: 8.0/10 CGPA.

Diamond Dril Sr. Sec. Public School, Greater Noida : — High School (10th standard) , 2021.Aggregate 78.6%;

Ram-Eesh International School, Greater Noida — Intermediate (12th standard) , 2023. Aggregate 73.4%;

03. MASTER SKILLS INVENTORY

Programming	C++, C, Python, JavaScript, SQL
Web	HTML5, CSS3, Tailwind CSS, responsive web development
Backend	FastAPI, REST APIs, AsyncIO, HTTPX, Pydantic
Databases	PostgreSQL, MySQL, MongoDB, Redis, Firebase Realtime Database
Cloud	AWS, Google Cloud (GCP), Firebase, Vercel, Render
DevOps/System	Linux commands, Git, GitHub, Docker, CI/CD concepts
Data Science	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Jupyter
BI/Analytics	Power BI, DAX, Snowflake, dbt, Microsoft Fabric, BigQuery
AI/GenAI	LLMs, RAG, LangChain, LangGraph, ChromaDB, vector embeddings, Hugging Face embeddings, Gemini API, Mistral AI
Other	Cloudinary, Playwright, BeautifulSoup, NetworkX, React Flow, Figma, Postman
Core CS	DSA, OOP, DBMS, OS, Computer Networks, SDLC, debugging, API integration

04. PROGRAMMING & CORE CS

C++: primary DSA language. Practice spans arrays, strings, linked lists, trees, graphs, recursion, backtracking, dynamic programming, stacks/queues, greedy and binary search.

Python: backend, APIs, data science, ML and AI/GenAI. **JavaScript:** web applications. **SQL:** databases and analytics.

Core CS interview areas: DSA, OOP, DBMS, OS, Computer Networks and SDLC. User has reported 300+ DSA problems on LeetCode and 200+ SQL/DB problems on CodeChef.

05. DATA SCIENCE & ANALYTICS

NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, SQL, Power BI and DAX.

Data engineering/analytics: Snowflake, dbt, Microsoft Fabric, BigQuery and ETL workflows.

ML work: Logistic Regression, Random Forest, SARIMAX, PCA/KMeans, feature engineering and model evaluation.

Customer analytics: custom synthetic retail dataset, SQL master dataset, EDA, segmentation, churn prediction, daily/monthly forecasting and a six-page Power BI dashboard. Earlier resume notes reported Logistic Regression accuracy ~0.8333 and Random Forest ~0.9167; SARIMAX for forecasting.

06. WEB & BACKEND

Frontend: HTML5, CSS, Tailwind CSS, JavaScript and responsive UI.

Backend: FastAPI, REST, AsyncIO, HTTPX and Pydantic.

Practical engineering: authentication, role-based workflows, realtime synchronization, file uploads, external API integration, structured logging, testing/debugging and Docker deployment.

07. DATABASES

Relational: PostgreSQL, MySQL. NoSQL/realtime: MongoDB, Redis, Firebase Realtime Database. Vector: ChromaDB. Cloud/data platforms: Snowflake, BigQuery, Microsoft Fabric. File storage: Firebase Storage/Cloudinary.

08. CLOUD / DEVOPS / DATA ENGINEERING

AWS: confirmed hands-on exposure/work. Exact service names are not asserted unless found in a source.

Google Cloud: confirmed hands-on exposure/work. Exact service names are not asserted unless found in a source.

Linux: learned Linux commands and command-line workflows.

Docker: used in project/deployment workflows, especially Swasthya AI CORE.

CI/CD: concepts are part of the engineering inventory; exact provider/pipeline not verified.

Data engineering: Airflow, Snowflake, dbt, Microsoft Fabric and BigQuery appear across project/learning material. Airflow debugging included HttpHook/PostgresHook and Docker issues.

09. AI / ML / GENAI

LLMs, RAG, LangChain, LangGraph, prompt engineering, vector embeddings, ChromaDB, Hugging Face embeddings, Gemini API and Mistral AI.

RAG concepts: ingestion, chunking, embeddings, semantic retrieval, MMR search, contextual memory, session-aware document handling and hallucination reduction.

SPECOPS-AI: active academic capstone concept focused on autonomous adversarial testing of enterprise LLM/GenAI applications, especially prompt injection and data leakage; not to be presented as a finished production project.

10. PROJECT REGISTRY — DEPLOYMENT / REPOSITORY / EVIDENCE LINKS

P01 — Swasthya AI CORE

Date: Jun 2026 – Jul 2026

Problem / purpose: Healthcare intelligence backend; REST APIs, async workflows, external provider aggregation, normalization/ranking, provider failover, structured logging and Docker deployment.

Technologies: Python, FastAPI, Redis, AsyncIO, HTTPX, Playwright, BeautifulSoup, Docker, Gemini API, Mistral AI, Render

Deployment URL: <https://swasthya-ai-core.onrender.com/>

Repository URL: <https://github.com/vibieprince/Swasthya-AI-CORE>

P02 — CourseMate AI

Date: Apr 2026 – May 2026

Problem / purpose: RAG-based PDF learning platform; PDF ingestion, chunking, embeddings, semantic/MMR retrieval, contextual memory, authentication and session-aware document handling.

Technologies: FastAPI, LangChain, ChromaDB, Hugging Face Embeddings, Mistral AI, Gemini API, Firebase, Python, JavaScript

Deployment URL: <https://vibieprince.github.io/CourseMate-AI>

Repository URL: <https://github.com/vibieprince/CourseMate-AI>

P03 — FareCast — Predictive Pricing & Fare Analytics

Date: Mar 2026 – Apr 2026

Problem / purpose: ML-powered fare prediction, real-time FastAPI REST services, traffic/weather/distance/temporal features, forecasting and booking-window recommendations.

Technologies: Python, FastAPI, TensorFlow, Scikit-learn, REST APIs, Firebase, Figma; historical resume also mentions TypeScript/HTML/CSS

Deployment URL: <https://vibieprince.github.io/FareCast>

Repository URL: <https://github.com/vibieprince/FareCast>

P04 — E-Commerce Supply Chain Analytics

Date: Jan 2026 – Feb 2026

Problem / purpose: Enterprise-style analytics pipeline using raw transactional data, dbt transformations, automated data-quality checks, K-Means customer segmentation and Power BI dashboards including Customer 360 and Risk & Loss.

Technologies: Microsoft Fabric Lakehouse/Warehouse, SQL, Python, Power BI, dbt, DAX, Scikit-learn

Deployment URL: <https://ecommercebyprince.vercel.app/>

Repository URL: <https://github.com/vibieprince/E-Commerce-Supply-Chain-Analytics-Project>

P05 — PrintSmart — College Print Order & Pickup System

Date: Jul 2025 – Dec 2025

Problem / purpose: Cloud web application for document upload, print configuration, UPI payment proof, order placement and realtime pickup/status tracking; Firebase authentication and Cloudinary file handling.

Technologies: HTML5, Tailwind CSS, JavaScript, Firebase, Cloudinary, Vercel

Deployment URL: <https://print-smart.vercel.app/>

Repository URL: <https://github.com/vibieprince/PrintSmart>

P06 — Sales Analytics Project using dbt

Date: Dec 2025 – Jan 2026

Problem / purpose: End-to-end e-commerce analytics platform; Snowflake RAW/STAGING/ANALYTICS layers, dbt transformations, star schema, data-quality tests and Power BI dashboards.

Technologies: Snowflake, Power BI, dbt, SQL, DAX

Deployment URL: [BLANK — not verified in accessible sources]

Repository URL: <https://github.com/vibieprince/SALES-ANALYTICS-DBT>

P07 — Discount Impact on Sales Volume in E-commerce

Date: Nov 2025 – Dec 2025

Problem / purpose: Analyzed discount effects on sales volume and revenue using EDA, correlation analysis and visualization; produced business-oriented recommendations/dashboard.

Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI, Jupyter Notebook

Deployment URL: [BLANK — not verified in accessible sources]

Repository URL: <https://github.com/vibieprince/Discount-Impact-on-Sales-Volume-in-E-commerce>

P08 — Customer Behavior, Sales Forecasting & Churn Prediction

Date: Nov 2025 – Dec 2025

Problem / purpose: Retail analytics pipeline built from custom realistic synthetic data; SQL master dataset, EDA, PCA/KMeans segmentation, churn model, daily/monthly forecasting and six-page Power BI dashboard.

Technologies: Python, SQL, Pandas, NumPy, Scikit-learn, Power BI; SARIMAX

Deployment URL: [BLANK — not verified in accessible sources]

Repository URL: <https://github.com/vibieprince/Customer-Behavior--Sales-Forecasting---Churn-Prediction-Platform>

P09 — RentEase / RentEase AI

Date: 2026

Problem / purpose: Rental property management system covering buildings, floors, rooms, tenants, meter readings, monthly billing and payment reconciliation. AI version uses LangGraph, Gemini and Google Sheets with deterministic financial services.

Technologies: Python, LangGraph, Gemini, Google Sheets API, Streamlit, Decimal, pytest; earlier version HTML/Tailwind/Vanilla JS/Firebase

Deployment URL: [BLANK — not verified in accessible sources]

Repository URL: <https://github.com/vibieprince/RENTEASE-AI>

Problem / purpose: Financial transaction graph/forensics concept for identifying suspicious relationships and visualizing transaction networks.

Technologies: NextJS, NetworkX, React Flow, Tailwind CSS

Deployment URL: [BLANK — not verified in accessible sources]

Repository URL: [BLANK — not verified in accessible sources]

Deployment URL: [BLANK — not verified in accessible sources]

Repository URL: [BLANK — not verified in accessible sources]

Deployment URL: [BLANK — not verified in accessible sources]

Repository URL: [BLANK — not verified in accessible sources]

Deployment URL: [BLANK — not verified in accessible sources]

Repository URL: [BLANK — not verified in accessible sources]

Deployment URL:

Repository URL:

***Deployment-link rule:** blank deployment fields are intentional. They are not placeholders to invent later; they mean no verified deployment URL was available in the accessible resume/public evidence. The two verified public deployment-style links found in resume material are CourseMate AI and FareCast GitHub Pages.*

11. HACKATHONS & COMPETITIVE BUILDS

- Code for Bharat Season 2 2025 — 2nd Runner-Up, AI Track, MITR SOS Device.
- HackSmart — BatterySmart × AWS Hackathon, Gurugram — Finalist.
- Smart India Hackathon / KrishiSetu AI / Kisan Guard — solution framing, slides and infographics.

- RIFT — financial forensics / transaction graph concept.

12. ACHIEVEMENTS

- 2nd Runner-Up — Code for Bharat Season 2 2025 (AI Track).
- HackSmart Finalist — BatterySmart × AWS Hackathon, Gurugram.
- 300+ DSA problems reported on LeetCode; public profile snapshot retrieved during this research showed 413 C++ problems, 69 PostgreSQL problems and 11 C problems.
- 200+ SQL/DB problems reported on CodeChef.
- Microsoft Certified: Power BI Data Analyst Associate (NASSCOM).

13. CERTIFICATIONS — CONSOLIDATED ACROSS RESUMES + PUBLIC PROFILE

C01 — Microsoft Certified: Power BI Data Analyst Associate

Issuer: Microsoft / NASSCOM | **Evidence:** Resume evidence: yes

Credential URL: https://drive.google.com/file/d/1pAhTS_DUXTfeJQC-VRM4Ns2Tq9IFzF45/view

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

C02 — HackerRank — SQL Advanced

Issuer: HackerRank | **Evidence:** Resume evidence: yes

Credential URL: <https://www.hackerrank.com/certificates/ce5279b0812d>

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

C03 — Oracle Certified — Agentic AI Foundations Associate

Issuer: Oracle | **Evidence:** Latest resume evidence: yes

Credential URL: <https://drive.google.com/file/d/1PCq2JnA4U6TixBBlluxlcFlxs-vcfFEC/view>

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

C04 — Python 101 for Data Science

Issuer: IBM | **Evidence:** Resume + LinkedIn evidence; Credential ID d58173cfb5544dcebace05192602f34

Credential URL: <https://courses.cognitiveclass.ai/certificates/d58173cfb5544dcebace05192602f34#>

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

C05 — Introduction to MongoDB

Issuer: MongoDB University | **Evidence:** Resume + LinkedIn evidence; Credential ID MDBx7rbo61evh

Credential URL: https://drive.google.com/file/d/1M5up6-77YIP2VYZk__IDC_kOs-J5SU6i/view

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

C06 — Apply AI: Analyze Customer Reviews

Issuer: Cisco Networking Academy | **Evidence:** Public LinkedIn evidence; issued Dec 2025

Credential URL: https://drive.google.com/file/d/1P1P8wU_IBblbHC0tVzZsHbtNnrQNYf2Q/view

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

C07 — Data Analytics Essentials

Issuer: Cisco Networking Academy | **Evidence:** Resume evidence: yes

Credential URL: <https://drive.google.com/file/d/1H2LArOmcSyXwD6FLMiMjrancxGCqWaG/view>

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

C08 — 30 Day SQL Micro Course

Issuer: Skill Course | **Evidence:** Public LinkedIn evidence; issued Nov 2025; Credential ID SQ21T01

Credential URL: <https://certificate.skillcourse.in/wp-content/uploads/qsm-certificates/140-12-23927d67433e8ac4196fa4dfdd48e411-0-0.pdf>

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

C09 — Microsoft Powered: Data Analyst 101

Issuer: Simplilearn | Evidence: Public LinkedIn evidence; issued Nov 2025; Credential ID 9497445

Credential URL: https://certificates.simplipdn.net/share/9497445_97782091764473028750.png

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

C10 — Intro to SQL

Issuer: Kaggle | Evidence: Public LinkedIn evidence; issued Aug 2025

Credential URL: [BLANK]

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

C11 — Data Cleaning

Issuer: Kaggle | Evidence: Public LinkedIn evidence; issued Apr 2025

Credential URL: [BLANK]

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

C12 — Brewcode 3.0 — Certificate of Participation

Issuer: Maharishi University of Information Technology | Evidence: Public LinkedIn evidence; issued Mar 2025

Credential URL: [BLANK]

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

C13 — Certificate of Participation in CodeWizard'25

Issuer: SRM Institute of Science and Technology | Evidence: Public LinkedIn evidence; issued Jan 2025; Credential ID 3bb5cfd4-796e-48ae-9652-9e01d851cc34

Credential URL: [BLANK]

Profile evidence: [linkedin.com/in/vibieprince](https://www.linkedin.com/in/vibieprince)

Credential-link rule: exact 'See credential' URLs were not exposed by the accessible LinkedIn result. They are intentionally left blank rather than fabricated. The LinkedIn profile is retained as the evidence link.

14. EXPERIENCE / PRACTICAL ENGINEERING

The available source set primarily documents project-based engineering, hackathons, academic work and self-directed development rather than a conventional long-term corporate employment history.

Engineering pattern: problem definition → architecture → implementation → integrations → testing/debugging → deployment/iteration.

Do not infer formal employment from projects or certifications.

15. LEADERSHIP / COLLABORATION

- Quick team integration and rapport-building; user describes ability to 'vibe with everyone'.
- Team/hackathon participation and collaborative academic work.
- Self-identified weakness: pushing through project errors too quickly; improvement strategy is systematic debugging and deliberate pauses before changing code.

16. GITHUB / OPEN SOURCE

Primary GitHub: github.com/vibieprince

Verified repositories: [Swasthya-AI-CORE](#) | [CourseMate-AI](#) | [RENTEASE-AI](#) | [E-Commerce-Supply-Chain-Analytics-Project](#)

Repository evidence should outrank generic profile text when answering implementation questions.

17. LEETCODE / CODECHEF / HACKERRANK

LeetCode: leetcode.com/u/vibieprince/

Public snapshot: username vibieprince; 413 C++ problems, 69 PostgreSQL problems, 11 C problems; 7 badges shown including 100 Days Badge 2025 and 50 Days Badge 2025/2026.

CodeChef: 200+ SQL/DB problems reported; exact profile URL not verified.

HackerRank: SQL Advanced certification documented; exact profile URL not verified.

18. CAREER INTERESTS

- Software/backend engineering.
- Cloud engineering / DevOps.
- Data science / analytics.
- AI / GenAI engineering.
- Data engineering.
- Early-career placement and building practical, deployable systems.

19. SOCIAL / CONTACT / PROFESSIONAL LINKS

Portfolio: <https://vibieprince.vercel.app>

GitHub: <https://github.com/vibieprince>

LinkedIn: <https://www.linkedin.com/in/vibieprince>

LeetCode: <https://leetcode.com/u/vibieprince/>

Instagram: https://www.instagram.com/reelify_prince

Email: prince.devds@gmail.com

Phone: +91 9560730326

Known handles: GitHub/LeetCode — vibieprince; Instagram — reelify_prince.

20. FAQ

Q: Who is Prince Singh?

A: B.Tech CSE (Data Science), GL Bajaj, 2023–2027; software/backend + data/AI + cloud-oriented profile.

Q: What are Prince's strongest technical areas?

A: Python/FastAPI/backend, C++/DSA, SQL/data analytics, RAG/GenAI, Docker, Linux, AWS and GCP.

Q: What is his best backend project?

A: Swasthya AI CORE.

Q: What is CourseMate AI?

A: A RAG PDF-learning system using FastAPI, LangChain, ChromaDB, embeddings, Mistral/Gemini and Firebase.

Q: What is PrintSmart?

A: A Firebase/Cloudinary web app for campus print ordering and realtime pickup tracking.

Q: What is FareCast?

A: An ML-powered fare prediction and forecasting platform with FastAPI REST APIs.

Q: Does he know AWS?

A: Yes, explicitly confirmed by the user and present in multiple resume versions.

Q: Does he know GCP?

A: Yes, explicitly confirmed by the user.

Q: Does he know Linux?

A: Yes, explicitly confirmed by the user.

Q: What certifications does he have?

A: See Section 13; the consolidated list includes Power BI, SQL Advanced, Oracle Agentic AI Foundations, IBM Python 101, MongoDB, Cisco AI/Data Analytics, Skill Course SQL, Simplilearn Data Analyst 101, Kaggle SQL/Data Cleaning and participation certificates.

Q: What are his major achievements?

A: Code for Bharat 2nd Runner-Up, HackSmart finalist, extensive DSA/SQL practice and technical certifications.

Q: Where is his portfolio?

A: <https://vibieprince.vercel.app>

Q: Where is his GitHub?

A: <https://github.com/vibieprince>

Q: Where is his LinkedIn?

A: <https://www.linkedin.com/in/vibieprince>

Q: Where is his LeetCode?

A: <https://leetcode.com/u/vibieprince/>

21. RAG GROUNDING / SOURCE RULES

- Treat explicit user-confirmed facts as high-confidence current facts.
- Treat current public profile information as current-profile evidence and retain the source/date.
- Treat historical resumes as historical snapshots; do not silently overwrite current facts.
- Never invent deployment URLs. Blank means unverified/unavailable.
- Never invent credential URLs. Blank means exact credential URL not exposed/verified.
- Distinguish project, hackathon, certification and employment.
- Do not convert a technology mentioned in a historical project into current expertise without confirmation.
- Use project records for project-specific questions and certification records for credential questions.
- When sources conflict, report the conflict and identify the source rather than choosing silently.
- Prefer repository evidence over resume summaries for implementation-level questions.

22. VERIFICATION & MISSING-DATA REGISTER

Portfolio pages: live Vercel portfolio could not be fetched reliably in this pass. A source export/ZIP is required for an exhaustive page-by-page crawl.

Deployment URLs: CourseMate AI and FareCast have verified GitHub Pages URLs in older resume evidence. Other project deployment fields are intentionally blank.

Certification URLs: LinkedIn exposes credential metadata but the exact 'See credential' URLs were not available in the retrieved result. Exact credential URLs are therefore blank.

Profile links: portfolio, GitHub, LinkedIn, LeetCode and Instagram are included. Exact CodeChef/HackerRank/Kaggle URLs remain unverified.

Current public-profile additions: LinkedIn lists Cisco Apply AI, 30 Day SQL Micro Course, Microsoft Powered Data Analyst 101, Kaggle Intro to SQL, Kaggle Data Cleaning, Brewcode 3.0 and CodeWizard'25 in addition to certifications present in several resumes.

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